Meaning of the spectral norm of a matrix

Asked 12 years, 4 months ago Modified 1 year, 9 months ago Viewed 65k times

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Is there an intuitive meaning for the spectral norm of a matrix? Why would an algorithm calculate the relative recovery in spectral norm between two images (i.e. one before the algorithm and the other after)? Thanks

matrices matrix-norms spectral-norm

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edited Aug 21, 2018 at 11:12
Rodrigo de Azevedo
1

asked Aug 29, 2012 at 3:57
val
1,037 1 11 18

2 Answers

Sorted by: Highest score (default)

76

The spectral norm (also know as Induced 2-norm) is the maximum singular value of a matrix. Intuitively, you can think of it as the maximum 'scale', by which the matrix can 'stretch' a vector.

The maximum singular value is the square root of the maximum eigenvalue or the maximum eigenvalue if the matrix is symmetric/hermitian

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edited Mar 10, 2023 at 23:05
VMMF
107 5

answered Aug 29, 2012 at 4:21
chaohuang
6,459 3 32 44

1 thanks - this helps me conceptually. – val Aug 29, 2012 at 6:56

6 This is specifically important since noise will be amplified by this value. For a very nice and intuitive understanding of this I recommend slides 21 and 22 of ee263.stanford.edu/lectures.html . – divB May 4, 2016 at 7:35

@divB, of which lecture? – jds Oct 10, 2018 at 0:29

3 27, "SVD and applications" – divB Oct 10, 2018 at 22:03

2 This is only true for symmetric matrices. math.stackexchange.com/questions/1437569/... – information_interchange Mar 28, 2020 at 2:09

31

Let us consider the singular value decomposition (SVD) of a matrix $X = USV^T$, where U and V are matrices containing the left and right singular vectors of X in their columns. S is a diagonal matrix containing the singular values. A intuitive way to think of the norm of X is in terms of the norm of the singular value vector in the diagonal of S. This is because the singular values measure the *energy* of the matrix in various principal directions.

One can now extend the p-norm for a finite-dimensional vector to a $m \times n$ matrix by working on this singular value vector:

$$\|X\|_p = \left(\sum_{i=1}^{\min(m,n)} \sigma_i^p\right)^{1/p}$$

This is called the *Schatten norm* of X. Specific choices of p yield commonly used matrix norms:

1. $p = 0$: Gives the rank of the matrix (number of non-zero singular values).
2. $p = 1$: Gives the nuclear norm (sum of absolute singular values). This is the tightest convex relaxation of the rank.
3. $p = 2$: Gives the Frobenius norm (square root of the sum of squares of singular values).
4. $p = \infty$: Gives the spectral norm (max. singular value).

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edited Mar 30, 2021 at 14:47
Rodrigo de Azevedo
1

answered Aug 29, 2012 at 6:07
Kartik Audhkhasi
1,566 9 12

1 thanks - can we really talk about Schatten p=0 for matrices? since we would be looking at 1/0...? – val Aug 29, 2012 at 7:02

1 Yes we can. And it is equal to the rank of the matrix. Can you clarify your question a bit more? – Kartik Audhkhasi Aug 29, 2012 at 13:41

2 Hi - i was referring to the Schatten norm equation above: the exponent is 1/p. If p=0 we have 1/0. thanks.. p must > 1. – val Aug 29, 2012 at 17:31

5 Think of $p \to 0$. Then any nonzero singular values will lead to 1, while 0 singular values will give 0 anyway. Hence, you will end up with the number of non-zero singular values. However, for $p \in [0, 1]$, the Schatten norm is not a "norm" since it does not satisfy all the properties. However, it is still common practice to call it a "norm". – Kartik Audhkhasi Aug 29, 2012 at 18:10

6 I just want to point out the confusion in your notation, the same notation: $||A||_2$ is also being used as spectral norm of a Matrix, which is the $p = \infty$ in your answer. I don't know what is Schatten Norm but one thing is universally agreed is that, matrix is an operator, and its norm should be defined in an operator fashion. – ArtificiallyIntelligent Nov 15, 2018 at 19:26